

PARIKSHA365 — STUDY NOTES

Geography — Common Book

SSC, RRB, Banks (GA), State PSC — Physical,
Indian, World, Human

SSC

RRB

BANKS

STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

EXAMINER BLUEPRINT — What Gets Tested Most

Geography in competitive exams splits into two zones: **physical geography** (earth's structure, atmosphere, rivers, climate, soils) and **Indian geography** (states, rivers, national parks, minerals, ports, passes). Examiners lean heavily on Indian geography, where factual precision matters most.

⚠ EXAM ALERT

****Top 8 geography exam hotspots****

1. ****India's rivers**** — which rivers flow east vs west, which originate in which mountain range, which drain into which sea, important dams. Appears in every SSC, RRB, and Banking exam.
2. ****States: superlatives**** — largest/smallest by area and population, longest coastline, wettest/driest, highest peak. Classic 1-mark questions.
3. ****National Parks and Wildlife Sanctuaries**** — first NP, only floating NP, Project Tiger reserves, UNESCO sites. Examiners love these.
4. ****Climate and Monsoon**** — types of monsoon, winter rain (NE monsoon vs Western disturbances), Cherrapunji vs Mawsynram.
5. ****Soils of India**** — black soil (cotton), red soil, laterite soil (not fertile), alluvial soil (most fertile, most extensive).
6. ****Minerals and their states**** — coal (Jharkhand, Odisha, WB), iron ore (Odisha, Jharkhand, Karnataka), mica (Rajasthan), bauxite (Odisha), petroleum (Assam, Gujarat, Rajasthan, Mumbai High).
7. ****Passes**** — Khyber, Bolan, Zoji La, Nathu La, Shipki La — which connects which.
8. ****World geography superlatives**** — longest river, highest peak, largest desert, deepest lake, largest ocean. Guaranteed 2–3 marks.

✓ KEY FACT

****Study priority for time-poor students****

1. India's rivers — east vs west flowing, origins, dams
2. State superlatives (area, population, coastline)
3. Soils of India — 4 types + which crop grows in which
4. National Parks — first, largest, only floating, biosphere reserves
5. Minerals — where each major mineral is found
6. Monsoon — branches, onset dates, withdrawal dates
7. World superlatives — 10 key facts
8. Passes — 5–6 key ones connecting India to neighbours

How To Use This Book

Geography rewards **visual memory** above all. You cannot remember the monsoon without a mental picture of the Bay of Bengal branch curving up into Assam; you cannot remember the southern peninsula's rivers without a map in your head with Godavari flowing east and Narmada flowing west. Build that mental map — the rest follows.

Keep a physical map of India visible while reading this book. Trace each river, each state border, each mountain range with your finger as you read about it. The physical act of tracing locks location into memory far better than reading alone.

This book moves **outward → inward**:

- **Part A — The Planet** — universe, solar system, earth's interior, rocks.
- **Part B — Forces Shaping Earth** — plate tectonics, earthquakes, volcanoes, atmosphere, climate.
- **Part C — Water + Biomes** — oceans, currents, tides, biomes, forests.
- **Part D — India, physical** — location, physiography, rivers, climate + monsoon, soils, natural vegetation, wildlife.
- **Part E — India, human + economic** — population, agriculture, industry, minerals + energy, transport, communication.
- **Part F — World, political + economic** — continents, countries, major cities, important resources, trade + organisations.

***Map rule:** keep a wall map of India + a world map visible while you read. Trace each feature with your finger. The kinaesthetic act locks location into memory much better than eye-reading alone.*

Part A — The Planet

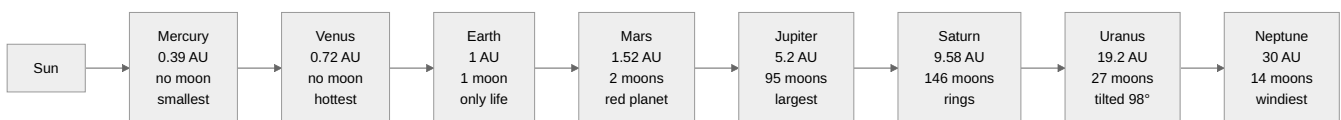
Chapter 1 — The Universe + the Solar System

1.1 Hook

13.8 billion years ago, all matter + energy was compressed into a point of infinite density — the **Big Bang**. Space-time expanded, hydrogen + helium coalesced into stars, stars cooked heavier elements + exploded as supernovae, and from the remnants our **Solar System** formed **~4.6 billion years ago**. The **Earth** coalesced **4.54 billion years ago**. **Life** began ~3.8 billion years ago. **Modern humans** — **~200,000 years ago**.

Our Sun is a **G-type main-sequence star (G2V)**, ~4.6 billion years old, ~5 billion years still to go. It lies about **~27,000 light-years** from the centre of the **Milky Way galaxy**. The Milky Way + its neighbours are part of the **Local Group**, part of the **Virgo Supercluster**, part of the **Laniakea Supercluster**.

1.2 The 8 planets — order + traits



Mnemonic — "My Very Educated Mother Just Served Us Noodles": Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune.

Inner (terrestrial): Mercury, Venus, Earth, Mars — small, rocky, few moons. **Outer (Jovian):** Jupiter, Saturn, Uranus, Neptune — large, gaseous, many moons + rings.

Asteroid belt lies between Mars + Jupiter. **Kuiper Belt** beyond Neptune (Pluto — reclassified as dwarf planet, 2006). **Oort Cloud** — source of long-period comets.

1.3 Earth — the home planet

- **Diameter:** 12,742 km.
- **Mass:** 5.972×10^{24} kg.
- **Axial tilt:** **23.5°** — the reason for **seasons**.
- **Rotation period:** 23h 56m 4s (sidereal day).
- **Revolution period:** 365.25 days.
- **Composition:** 70.8% water surface (oceans + ice); 29.2% land.
- **Atmosphere:** 78% N₂ + 21% O₂ + 0.93% Ar + 0.04% CO₂ + trace.

1.4 The Moon + eclipses

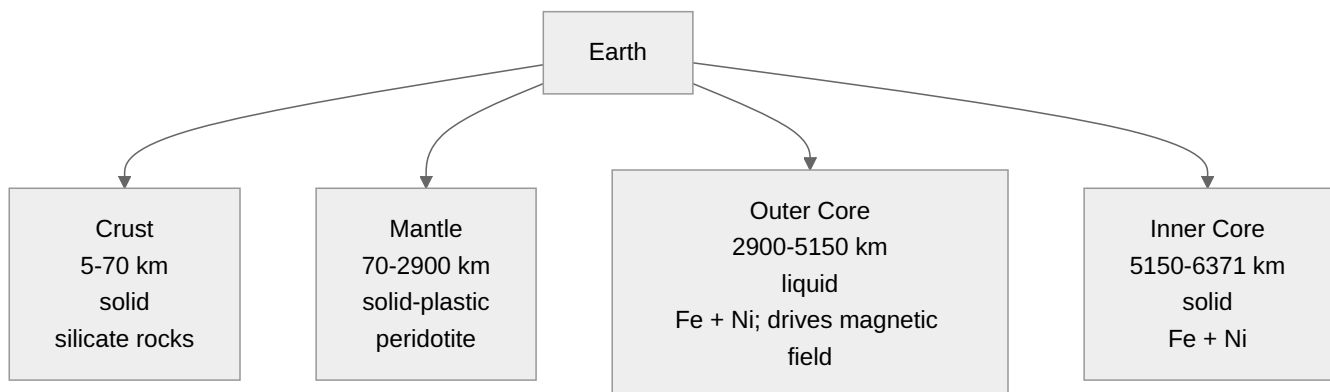
- Moon orbits earth at **~384,400 km**; period **27.3 days** (sidereal); synodic (new-to-new) **29.5 days**.
- Same face always visible — **tidally locked**.
- **Causes tides** via gravitational pull (lunar + solar tides; spring tides at new/full moon; neap tides at quarter moons).
- **Solar eclipse** — new moon passes between sun + earth (Sun shadowed). Types: total, annular ("ring of fire"), partial, hybrid.
- **Lunar eclipse** — full moon passes into earth's shadow (Moon reddens — "blood moon" via Rayleigh scattering).

1.5 Exam hooks

- Order of planets — MVEMJSUN mnemonic.
- **Mercury** smallest; **Jupiter** largest; **Venus** hottest (runaway greenhouse, ~462 °C); **Neptune** windiest.
- **Axial tilt 23.5°** → cause of seasons.
- Pluto reclassified → **dwarf planet (2006, IAU, Prague)**.
- Solar eclipse = **new moon**; Lunar eclipse = **full moon**.

Chapter 2 — Earth's Interior + Rocks

2.1 The 4 layers

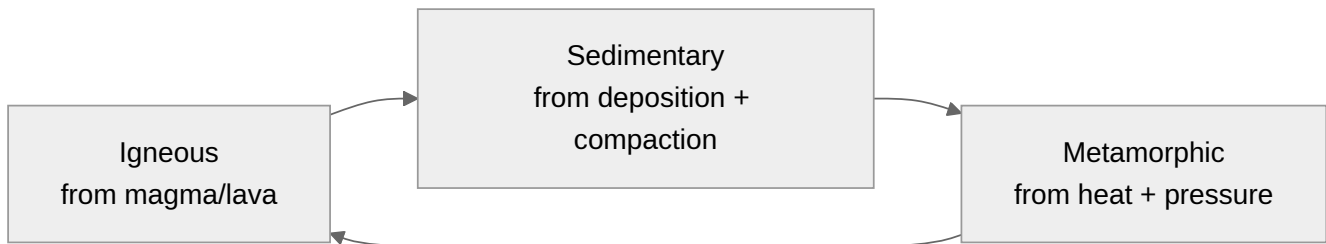


- **Crust** — continental crust avg 35 km thick; oceanic crust 5-10 km thin
- denser. Main rocks: granite (continental) + basalt (oceanic).
- **Mantle** — 84% of Earth's volume; solid but plastic at depth (flows over geological timescales); **asthenosphere** (~100–200 km depth) is where tectonic plates move.
- **Core** — outer liquid, inner solid; temperature ~5,200°C at inner core.

Discontinuities (named after their discoverers): - **Conrad** — within continental crust (granite → basalt). - **Mohorovičić (Moho)** — crust/mantle boundary. - **Gutenberg** — mantle/outer core. -

Lehmann — outer core/inner core. - **Repetti** — upper/lower mantle.

2.2 Rock cycle — 3 types



Igneous (from magma): - **Intrusive / plutonic** — cooled slowly underground; coarse grains (large crystals); e.g., **granite, gabbro**. - **Extrusive / volcanic** — cooled quickly on surface; fine grains; e.g., **basalt, rhyolite, pumice, obsidian**.

Sedimentary (layered; often fossil-bearing): - **Clastic** — from rock fragments (**sandstone, shale, conglomerate**). - **Organic** — from dead organisms (**coal, limestone from coral**). - **Chemical** — precipitation (**rock salt, gypsum**).

Metamorphic (heat + pressure change existing rocks): - **Granite → Gneiss - Shale → Slate → Phyllite → Schist - Limestone → Marble - Sandstone → Quartzite - Coal → Graphite → Diamond - Basalt → Amphibolite**

2.3 Exam hooks

- Moho = crust/mantle; Gutenberg = mantle/outer core; Lehmann = outer/inner core.
- **Basalt** = oceanic crust; **Granite** = continental crust.
- **Deccan Traps** = basalt from volcanic eruptions ~66 Mya (K-Pg).
- **Marble** = metamorphosed limestone.
- **Diamond** = metamorphosed coal (carbon, highest-grade).

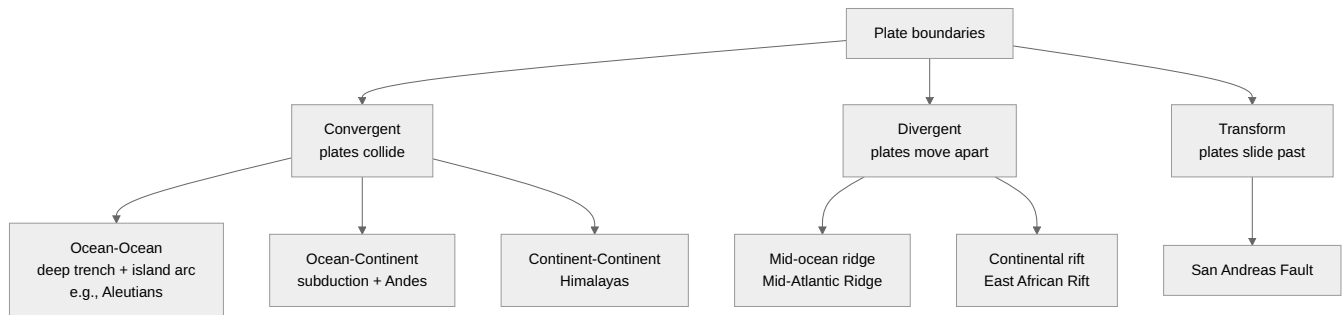
Chapter 3 — Plate Tectonics, Earthquakes, Volcanoes

3.1 The big idea

Earth's **lithosphere** (crust + upper mantle) is broken into **~15 major plates** that float on the plastic **asthenosphere**. They move a few cm/year — slow but inexorable. At their boundaries, we get **earthquakes, volcanoes, mountain-building, ocean trenches**.

Continental Drift hypothesis — **Alfred Wegener (1912)** — proposed all continents were once one — **Pangaea** — that drifted apart. Confirmed by mid-20th century with sea-floor spreading evidence (Harry Hess, 1960s) + palaeomagnetism (Vine-Matthews-Morley) → **Plate Tectonics theory** by 1968.

3.2 Types of plate boundaries



Himalayas — **Indian plate** colliding with **Eurasian plate** since ~50 million years ago; still rising ~5 mm/yr.

3.3 Earthquakes

- **Focus (hypocenter)** — point inside Earth where rupture starts.
- **Epicenter** — point on surface directly above focus.
- **Richter scale** (1935, Charles Richter) — logarithmic, open-ended; each step = 10× amplitude + ~32× energy.
- **Moment Magnitude (M_w)** — used by USGS now for M ≥ 5.
- **Mercalli scale** — intensity felt, qualitative I–XII.

Waves: - **P (primary)** — longitudinal, fastest, through solid + liquid. - **S (secondary)** — transverse, slower, through solids only (how we know outer core is liquid!). - **L (surface)** — slowest, most damaging.

World seismic zones: - **Circum-Pacific Belt** — "Ring of Fire" — ~81% of quakes. Japan, Philippines, Indonesia, W. Americas. - **Alpide Belt** — Mediterranean through Himalayas — 17%. - **Mid-Atlantic Ridge** — divergent, moderate.

India's 5 seismic zones (BIS, 2002 — simplified to 4): Zone II (low), III (moderate), IV (severe), V (very severe — NE, J&K, parts of Uttarakhand, Rann of Kutch, Andamans).

Major recent earthquakes in India: - **Bhuj, Gujarat** — 26 Jan 2001, M_w 7.7, ~20,000 deaths. - **Kashmir** — 8 Oct 2005, M_w 7.6, ~80,000 deaths (mostly PoK). - **Sikkim** — 18 Sep 2011, M_w 6.9. - **Nepal** (spillover) — 25 Apr 2015, M_w 7.8, ~9,000 deaths.

3.4 Volcanoes

- **Active** (erupted within recent past): Barren Island (Andamans — India's only active), Kilauea (Hawaii).
- **Dormant** (no recent eruption but could): Mt Vesuvius (Italy).
- **Extinct** (no expected future eruption): Mt Aconcagua (Argentina); Deccan Traps today.

Types — shield (gentle, Hawaiian, basalt, low viscosity), composite (stratovolcano, violent, Mt St Helens + Fuji), cinder cone (small pyroclastic), caldera (collapsed — Krakatoa, Toba).

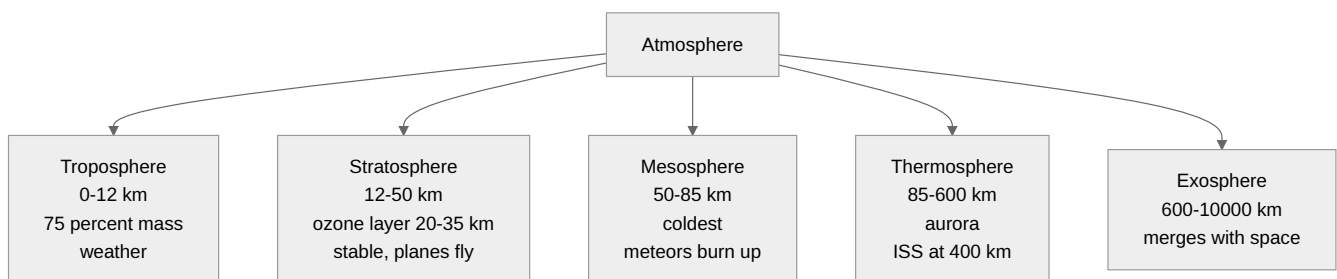
Deccan Traps — India's great volcanic province; **65–66 Mya** eruptions spewed ~500,000 km² of basalt; possibly contributed (alongside Chicxulub asteroid) to the **K-Pg mass extinction** that killed the non-avian dinosaurs.

3.5 Exam hooks

- Wegener (1912) — Continental Drift.
- Plate Tectonics formalised 1968.
- Indian plate colliding Eurasian plate → Himalayas (50 Mya, rising 5 mm/yr).
- **S-waves cannot travel through liquid** → outer core is liquid.
- **Ring of Fire** = 81% of quakes.
- **Barren Island** = India's only active volcano.
- **Mauna Kea** (Hawaii) is the tallest mountain if measured from seafloor.
- Mt Everest (8848 m) is the tallest above sea level; in **Nepal-China** border.
- **Krakatoa** eruption 1883 → one of the loudest sounds in history.

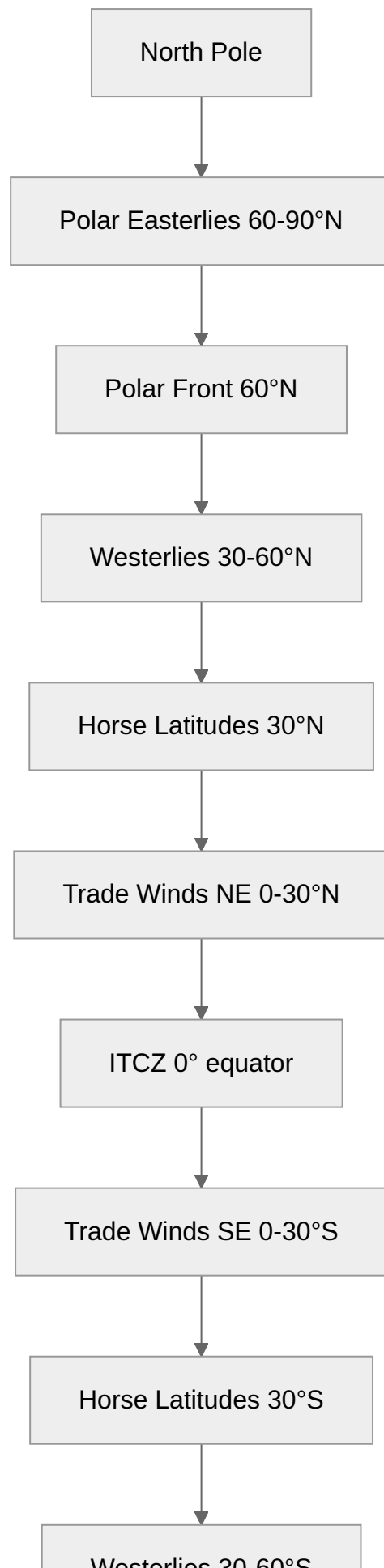
Chapter 4 — Atmosphere + Climate

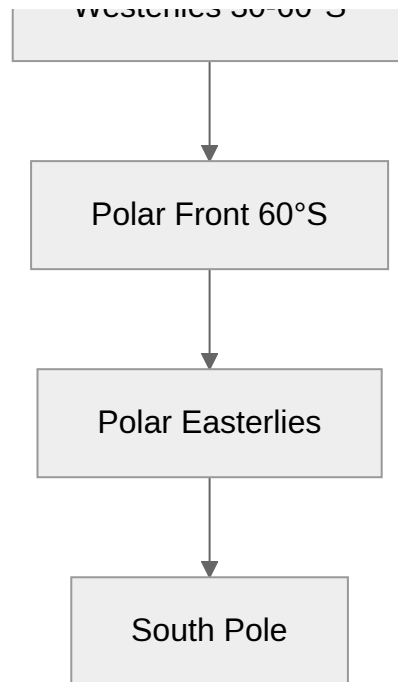
4.1 The 5 atmospheric layers



Kármán Line — 100 km = boundary of space (FAI). Temperature gradient: falls in troposphere (-6.5°C/km), rises in stratosphere (due to ozone absorbing UV), falls in mesosphere, rises in thermosphere.

4.2 Winds — the 3-cell model





Three circulation cells per hemisphere: Hadley (0–30°), Ferrel (30–60°), Polar (60–90°).

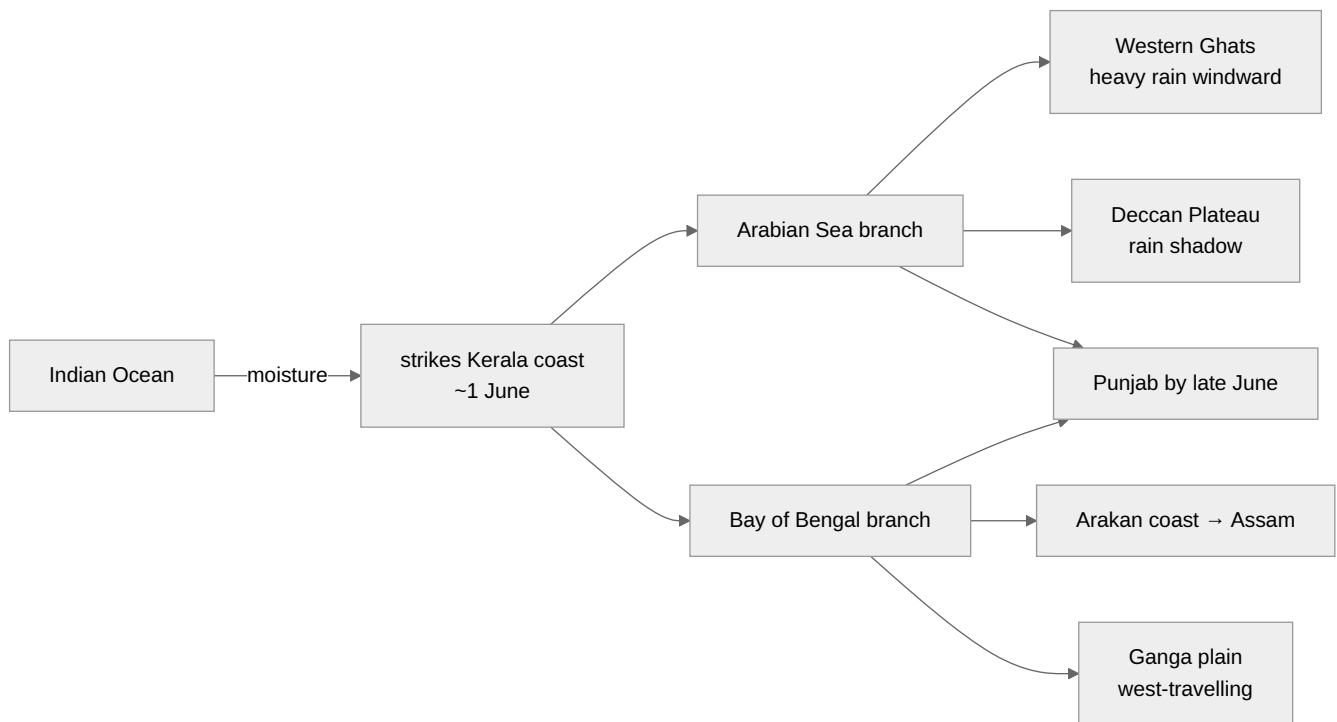
Coriolis Effect — Earth's rotation deflects winds **right** in NH, **left** in SH. Zero at equator; maximum at poles.

Named local winds: - **Loo** — hot, dry N India summer. - **Mango Showers / Blossom Showers** — pre-monsoon thunderstorms Kerala-KA. - **Nor'westers / Kaalbaisakhi** — pre-monsoon W. Bengal. - **Chinook** — "snow-eater", USA Rockies' lee. - **Foehn** — Alps' lee side. - **Sirocco** — hot dusty Mediterranean (N Africa origin). - **Mistral** — cold dry France, Rhone Valley. - **Harmattan** — dry W. Africa. - **Santa Ana** — California dry hot.

4.3 Indian Monsoon — the crown jewel topic

Mechanism — simple version: In summer (June–September), the **Indian subcontinent heats up** faster than the surrounding **Indian Ocean**. Hot air rises over land → low pressure forms. High-pressure oceanic air rushes in from the south → moisture-laden **SW monsoon winds**. In winter, the pattern reverses: cold land → high pressure; warm ocean → low pressure → **NE / retreating monsoon winds** from land to sea.

The 2 branches of SW monsoon:



Timeline: - **1 June** — Kerala onset (IMD declares). - **10 June** — Mumbai. - **15 June** — Kolkata, parts of NE. - **End June** — Delhi. - **15 July** — entire India covered. - **1 Sep onwards** — retreat begins NW → SE. - **15 Oct** — monsoon fully retreated.

NE monsoon (Oct–Dec) — brings ~60% of **Tamil Nadu's** annual rainfall; also coastal AP + Rayalaseema + Karaikal + S. Kerala. Rest of India is dry.

Mawsynram (Meghalaya) — **wettest place on Earth** (~11,872 mm avg). **Cherrapunji (Sohra)** next (~11,777 mm).

Drivers of variability: - **El Niño** (anomalous Pacific warming) → weakens Indian monsoon. - **La Niña** (Pacific cooling) → strengthens. - **Indian Ocean Dipole (IOD)** — positive phase (W. IO warmer than E.) strengthens monsoon; negative phase weakens. - **MJO** (Madden-Julian Oscillation) — eastward-moving tropical convection.

4.4 Exam hooks

- Troposphere has 75% of atmosphere's mass.
- **Ozone layer** is in **stratosphere**, 20–35 km.
- Kármán Line = **100 km**.
- **Coriolis** = zero at equator.
- **Loo** = N India hot dry wind; **Nor'westers** = W. Bengal; **Mango showers** = pre-monsoon Kerala.
- SW monsoon reaches Kerala **1 June**; retreats by 15 Oct.
- 2 branches: **Arabian Sea** + **Bay of Bengal**.
- Mawsynram = wettest place.
- **El Niño** weakens monsoon.
- Western Ghats = windward heavy rain; Deccan = rain shadow.

Chapter 5 — Oceans + Currents + Tides

5.1 Five oceans

Pacific > Atlantic > Indian > Southern (Antarctic) > Arctic.

- **Deepest point: Challenger Deep**, Mariana Trench, Pacific — ~10,984 m.
- **Largest ocean:** Pacific, ~46% of all ocean water.
- **Busiest trade ocean:** Atlantic.
- **Youngest ocean:** Atlantic (from break-up of Pangaea).
- **Southern Ocean** recognised as 5th by IHO (2000); NatGeo (2021).

5.2 Ocean currents — the conveyor belts

Warm currents move water from equator toward poles; **cold currents** from poles toward equator.

Ocean	Warm current	Cold current
Pacific	Kuroshio (Japan), Equatorial Counter, East Australian	California, Peru (Humboldt), Oyashio
Atlantic	Gulf Stream, N. Atlantic Drift, Brazil	Canary, Labrador, Benguela
Indian	Agulhas, Mozambique, Monsoon current (seasonal)	W. Australian

Gulf Stream — brings warm Gulf-of-Mexico water up the E. US coast, then across Atlantic as **N.**

Atlantic Drift — reason **W. Europe** is milder than its latitude suggests.

El Niño occurs in equatorial Pacific; **upwelling** of cold Humboldt current off Peru drives fisheries + rainfall patterns.

5.3 Tides

- Caused by **gravitational pull** of Moon (dominant) + Sun.
- **Spring tides** — Sun + Moon aligned (new/full moon) → highest range.
- **Neap tides** — Sun + Moon perpendicular (quarter moons) → lowest range.
- Usually 2 high + 2 low tides per day (semi-diurnal).

India's tidal stations: Kandla (Gulf of Kutch — highest tidal range in India, ~7 m); Okha, Mumbai, Haldia, Paradip, Chennai, Vizag, Cochin, Tuticorin.

5.4 Exam hooks

- Deepest = Challenger Deep (Mariana, Pacific), 10,984 m.
- Largest = Pacific.
- **Gulf Stream + N Atlantic Drift** warm W Europe.
- **Kuroshio** = Japan; **Humboldt (Peru)** = cold, fisheries.
- **Spring tide** = new/full moon.

- **Neap tide** = quarter moons.
-

Part D — India, Physical

Chapter 6 — India: Location + Extent + Borders

6.1 Key location facts

- **Latitudinal extent:** 8°4'N (Kanyakumari) to 37°6'N (Indira Col, Ladakh) = ~30° range.
- **Longitudinal extent:** 68°7'E (Guhar Moti, Kutch) to 97°25'E (Kibithu, Arunachal) = ~30° range.
- **Time difference:** ~2 hrs between eastern-most + western-most.
- **IST** = UTC + 5:30 hrs, based on **82°30'E longitude** passing through **Mirzapur, UP**.
- **Tropic of Cancer (23.5°N)** passes through **8 Indian states:** Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram. Mnemonic — "**Gujarat Rajasthan Ma CH JH BeTrMi**" → GR MaCH JH BeTrMi (or any 8-item jingle).
- **Area:** 3.287 million km² — **7th largest country**.
- **Coastline:** 7,516.6 km (mainland + islands).
- **Land border:** 15,106.7 km, shared with 7 countries.

6.2 Land borders (mnemonic — "BBC + MAP-N"):

Starting clockwise from W: - **Pakistan** (3,323 km, Radcliffe Line) — 4 states: Gujarat, Rajasthan, Punjab, J&K. - **Afghanistan** (106 km, **Durand Line**; entirely through PoK). - **China** (3,488 km, **McMahon Line** in E, LAC overall) — 5 states: Ladakh, HP, Uttarakhand, Sikkim, Arunachal. - **Nepal** (1,751 km) — 5 states: Uttarakhand, UP, Bihar, WB, Sikkim. - **Bhutan** (699 km) — 4 states: Sikkim, WB, Assam, Arunachal. - **Bangladesh** (4,096 km — longest) — 5 states: WB, Assam, Meghalaya, Tripura, Mizoram. - **Myanmar** (1,643 km) — 4 states: Arunachal, Nagaland, Manipur, Mizoram.

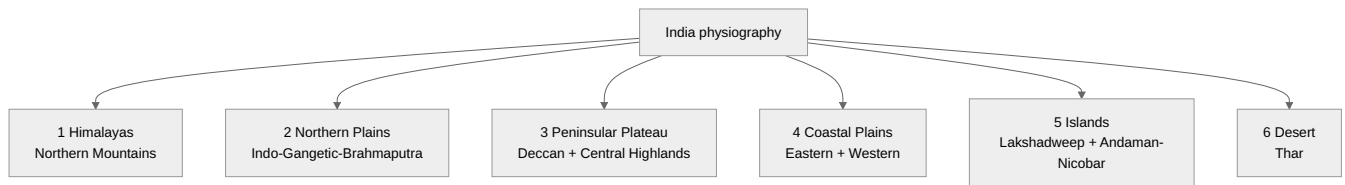
Maritime: Sri Lanka (**Palk Strait**), Maldives.

6.3 Exam hooks

- IST longitude = **82°30'E** (Mirzapur).
- Tropic of Cancer crosses **8 states**.
- **Bangladesh** = longest land border (4,096 km).
- **Afghanistan** = shortest (106 km, through PoK).
- **McMahon Line** = India-China (E); **Durand Line** = India-Afghanistan; **Radcliffe Line** = India-Pakistan.
- Standard Meridian — 1 state covers it (UP).
- Indira Point (Great Nicobar) = southernmost tip of Indian territory (overall); Kanyakumari = southernmost on mainland.

Chapter 7 — Physiography of India

Divided into **6 macro-regions**:



7.1 Himalayas — 3 parallel ranges + 4 divisions by longitude

By parallel range (S to N):

1. **Shiwaliks** (Outer Himalayas) — foothills, 900–1,200 m; Duns here (Dehra Dun, Kotli Dun).
2. **Lesser / Middle Himalayas** — Pir Panjal, Dhauladhar, Mahabharat, Mussoorie, Nag Tibba; avg 3,700–4,500 m; famous hill stations (Shimla, Mussoorie, Nainital, Darjeeling).
3. **Greater Himalayas / Himadri** — Mount Everest, Kanchenjunga, Nanda Devi, Nanga Parbat; perpetually snow-covered; 6,000+ m.
4. **Trans-Himalayas** — N of Greater, includes Karakoram, Ladakh, Zaskar; **K2 (8611 m)** = 2nd highest peak globally (in PoK Karakoram).

By longitude (W to E):

- **Punjab / Kashmir Himalayas** — Indus to Sutlej.
- **Kumaon Himalayas** — Sutlej to Kali.
- **Nepal Himalayas** — Kali to Teesta (Everest here).
- **Assam Himalayas** — Teesta to Brahmaputra (Arunachal).

Mount Everest — 8,848 m — India's nearest peak Kanchenjunga (8,586 m, 3rd in world) in **Sikkim** = India's highest.

Purvanchal Range — NE, extends from Arunachal to Mizoram: Patkai (Naga + Arunachal border), Naga Hills, Manipur Hills, Mizo Hills (Lushai).

7.2 Northern Plains — the Indo-Gangetic-Brahmaputra

- ~2,400 km long, 150–300 km wide; 7 lakh km²; world's most fertile + densely populated.
- 3 major rivers systems:
- **Indus** — W (Punjab + Sindh).
- **Ganga** — largest basin in India; Varanasi to Gangetic Delta.
- **Brahmaputra** — from Tibet (Tsangpo) through Arunachal (Dihang) to Assam + Bangladesh (Jamuna).
- **Bhabar** (piedmont) → **Tarai** (marshy) → **Bhangar** (old alluvium) → **Khadar** (new alluvium — most fertile).

7.3 Peninsular Plateau

India's oldest landmass. 2 divisions:

- **Central Highlands:** Aravallis + Vindhyas + Satpuras + Maikal + Kaimur; Malwa + Bundelkhand + Baghelkhand plateaus.
- **Deccan Plateau:** S of Narmada; flanked by **Western Ghats** (Sahyadri)
- **Eastern Ghats.** Rivers Godavari + Krishna + Kaveri flow east; Narmada + Tapi flow west (through rift valleys between Vindhyas + Satpuras).

Western Ghats: UNESCO WHS 2012; Biodiversity hotspot; **Anamudi (Kerala, 2,695 m)** = highest peak in peninsular India.

Eastern Ghats: broken, less continuous; highest peak **Arma Konda** (AP, 1,680 m). Meet Western Ghats at **Nilgiri Hills** in Tamil Nadu + Karnataka + Kerala (**Doddabetta** = Nilgiri's highest, 2,637 m).

7.4 Coastal plains

- **Eastern coast** — wider, deltas (Mahanadi, Godavari, Krishna, Kaveri); Coromandel Coast (TN) + Northern Circars (AP + Odisha).
- **Western coast** — narrower, estuaries (not deltas); divided into Konkan (Maha + Goa), Kanara (Karnataka), Malabar (Kerala). Kerala Backwaters here.

7.5 Islands

- **Lakshadweep** — 36 Arabian Sea islands, all coral atolls; capital **Kavaratti**; ~32 km² — India's smallest UT by area.
- **Andaman + Nicobar** — 572 islands, volcanic origin; capital **Port Blair** (renamed **Shri Vijaya Puram**, 2024); Port Blair is closer to Yangon (Myanmar) than to Chennai.

7.6 Thar Desert

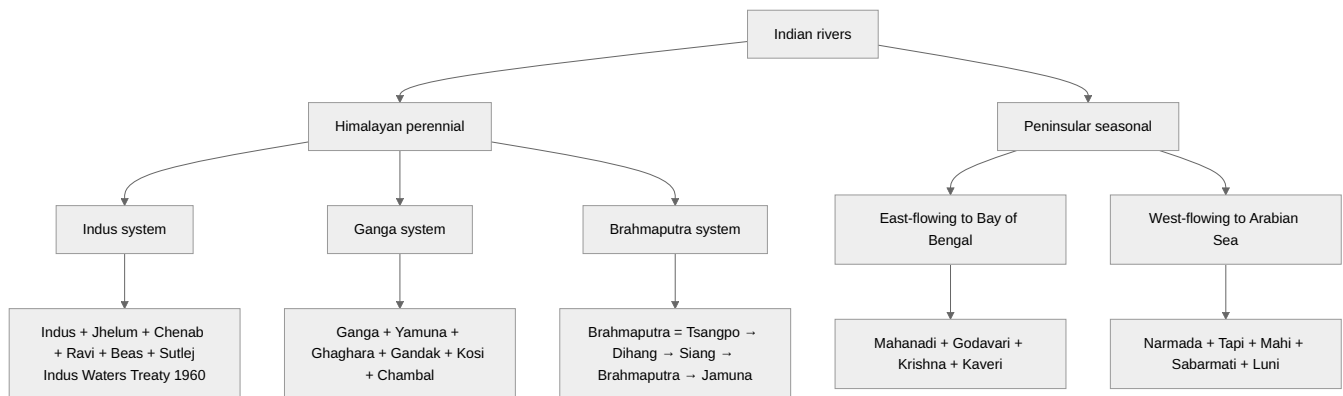
- In **W. Rajasthan** + extending into Gujarat + Pakistan.
- Luni is the only major river (seasonal, ephemeral).
- Indira Gandhi Canal (from Sutlej + Beas at Harike barrage) greening parts of Thar.

7.7 Exam hooks

- **Kanchenjunga** (8,586 m, Sikkim) = India's highest peak.
- **Anamudi** (Kerala, 2,695 m) = highest peak of peninsular India.
- Khadar = new alluvium; Bhangar = old alluvium.
- Narmada + Tapi flow **west** through rift valleys.
- Western Ghats = Sahyadri; Eastern Ghats = discontinuous.
- Doddabetta = Nilgiri's highest (2,637 m).
- Port Blair renamed **Shri Vijaya Puram** (2024).

Chapter 8 — Rivers of India

8.1 The 2 river systems



8.2 Himalayan rivers — key facts

- **Indus** — 3,180 km (710 km in India) — from Tibet; enters India at Leh; flows through J&K, Ladakh, PoK, Pakistan → Arabian Sea.
- **Ganga** — 2,525 km (largest basin in India) — **Source: Gangotri (Bhagirathi + Alakananda meet at Devprayag to become Ganga)**; Bay of Bengal via **Sundarbans delta** (world's largest).
- **Yamuna** — largest tributary of Ganga; origin Yamunotri (Uttarakhand). Flows past Delhi, Agra (Taj), Mathura, Prayagraj (Sangam with Ganga).
- **Brahmaputra** — 2,900 km total, 916 km in India; source **Manasarovar** (Tibet) as **Yarlung Tsangpo**; enters India in Arunachal as **Dihang/Siang**; joins with Dibang + Lohit at Sadiya to become Brahmaputra.
- **Kosi** — "Sorrow of Bihar" (shifts course); source Nepal Himalaya.
- **Ghaghara** — longest Himalayan tributary of Ganga; Nepal source (Karnali).

8.3 Peninsular rivers

× COMMON TRAP

****East-flowing vs West-flowing rivers — the classic trap**** The vast majority of peninsular rivers flow EAST into the Bay of Bengal (Godavari, Krishna, Cauvery, Mahanadi). Only a few flow WEST into the Arabian Sea. Remember the ****west-flowing exceptions****: Narmada, Tapi, Sabarmati, Mahi, Luni, Periyar. These flow through rift valleys between the Vindhyas and Satpuras (Narmada, Tapi) or through Gujarat (Sabarmati, Mahi). Since they flow through rift valleys (not normal river valleys), they form estuaries — not deltas. Mnemonic: *****"NTSML"***** = Narmada, Tapi, Sabarmati, Mahi, Luni — the five main west-flowing peninsular rivers.

East-flowing (to Bay of Bengal; deltas): - **Godavari** — 1,465 km, **longest Peninsular river + longest entirely within India**. Source: **Trimbak, Nashik** (MH). "Dakshina Ganga." - **Krishna** — 1,400 km. Source: **Mahabaleshwar** (MH). - **Kaveri** — 805 km. Source: **Talakaveri, Coorg** (KA). Disputes: TN-KA. - **Mahanadi** — 858 km. Source: Chhattisgarh. Hirakud Dam.

West-flowing (to Arabian Sea; estuaries, not deltas; through rift valleys): - **Narmada** — 1,312 km. Source: **Amarkantak** (MP). Flows W through rift between Vindhya + Satpura. Sardar Sarovar Dam. - **Tapi / Tapti** — 724 km. Source: **Multai** (MP). Similar rift pattern. - **Sabarmati** — 371 km (Gujarat; Dhebar Lake source). - **Mahi** — 583 km (MP → Guj). - **Luni** — Rajasthan; ephemeral; ends in Rann.

8.4 Indus Waters Treaty (1960)

Signed 19 September 1960 by Nehru + Ayub Khan + World Bank (mediator). Allocates: - **Eastern rivers** (Ravi, Beas, Sutlej — 20%) to **India** — unrestricted use. - **Western rivers** (Indus, Jhelum, Chenab — 80%) to **Pakistan** — India can use for domestic, non-consumptive, agriculture (limited), hydropower (run-of-river only).

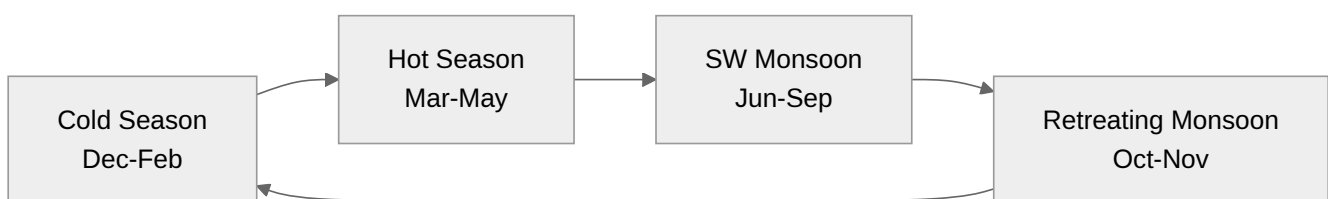
Most resilient cross-border water treaty — survived 3 wars.

8.5 Exam hooks

- Ganga source = **Gangotri / Devprayag**.
- Yamuna source = **Yamunotri**.
- Brahmaputra source = **Manasarovar**, Tibet.
- **Godavari** = longest peninsular + longest entirely in India.
- Narmada source = **Amarkantak**; Tapi source = **Multai** — both MP.
- Narmada + Tapi flow **WEST**.
- **Kosi** = Sorrow of Bihar; **Damodar** = Sorrow of Bengal (pre-dams).
- Indus Waters Treaty = **1960**.

Chapter 9 — Indian Climate + Monsoon

9.1 4 seasons



- **Cold (Dec–Feb)** — Western disturbances bring winter rain to N India (important for Rabi); clear skies elsewhere.
- **Hot (Mar–May)** — temperatures rise; pre-monsoon thunderstorms (Loo, Nor'westers, Mango showers).
- **SW Monsoon (Jun–Sep)** — gives **~75% of India's annual rainfall**.
- **Retreating / NE Monsoon (Oct–Nov)** — dry for most; wet for TN + AP coast.

(SW + NE monsoon mechanics — see Chapter 4 §4.3.)

9.2 Climate classification

Köppen's 6 classes in India:

1. **Aw** — Tropical savanna (most of central peninsula).
2. **Am** — Tropical monsoon (W. Ghats windward).
3. **Cwg** — Humid subtropical (N plains).
4. **Bwh** — Hot desert (Thar).
5. **Bsh** — Semi-arid (Rajasthan + parts of peninsular rain-shadow).
6. **ET / H** — Tundra / alpine (Himalayas).

9.3 Exam hooks

- SW monsoon = **75% of annual rain**.
- NE monsoon = most rain for **Tamil Nadu** + coastal AP.
- **Mawsynram** (Meghalaya) = wettest place on earth.
- Köppen most-common class in India = **Aw (tropical savanna)**.
- **Western Disturbances** = cold-season rain in N India.

Chapter 10 — Soils of India

6 major soil types (ICAR classification):

Soil	% of India	Region	Crops
Alluvial	~43%	Indo-Gangetic plain, coastal + delta	Rice, wheat, sugarcane, jute, oilseeds
Black (Regur)	~15%	Deccan Trap region — Maha, Guj, MP, KA, AP	Cotton , sugarcane, oilseeds
Red + Yellow	~18%	Peninsular (Deccan + parts of E peninsula)	Millets, groundnut, cotton
Laterite	—	Kerala, Karnataka, Odisha, Assam — high-rainfall hills	Cashew, tea, coffee, rubber
Arid + Desert	—	Rajasthan + parts of Gujarat	Barley, bajra (with irrigation)
Forest + Mountain	—	Himalayas	Tea, coffee, spices

Also: Saline / Alkaline (Reh — Punjab, Haryana); Peaty + Marshy (Kerala Kuttanad, Sundarbans).

10.1 Exam hooks

⚠ EXAM ALERT

****Soils of India — the 4-type table every exam asks about**** | Soil | Where found | Key crop | Special feature | ---|---|---|---| | ****Alluvial**** | Indo-Gangetic plains, river deltas, coastal plains | Wheat, rice, sugarcane | Most extensive + most fertile | | ****Black (Regur)**** | Deccan Plateau, Maharashtra, MP, Karnataka | ****Cotton**** | Self-ploughing (cracks in summer); derived from Deccan basalt; retains moisture | | ****Red and Yellow**** | Odisha, Chhattisgarh, Eastern MP, South India | Millets, groundnut | Iron oxide = red colour; porous, less fertile | | ****Laterite**** | Western Ghats slopes, NE India, Tamil Nadu hills | Tea, coffee, cashew | Heavy rainfall leaches nutrients; poor fertility; good for plantation crops | Khadar = new alluvium (lighter, sandy, near river banks) · Bhangar = old alluvium (darker, calcareous nodules, farther from river)

- **Alluvial** = most widespread + most fertile.
 - **Black soil** = best for **cotton**; self-ploughing (cracks in summer); derived from **Deccan basalt**.
 - **Laterite** = high-rainfall + temperature; poor fertility.
 - **Red soil** gets its colour from **iron oxide**.
 - Khadar (new alluvium) vs Bhangar (old alluvium).
-

Part E — India: Human + Economic

Chapter 11 — Population

- **India overtook China in April 2023** — most populous country, ~144 crore (2024).
- **17.5% of world population on 2.4% of land.**
- **TFR 2.0** (NFHS-5, 2019–21) — below replacement (2.1) for first time.
- **Sex ratio: 1,020 females per 1,000 males** (NFHS-5, first time above 1000 in Indian history); **Kerala** best (1,084); **Haryana** worst (879).
- **Median age:** ~28 years → **demographic dividend.**
- **Literacy:** 77.7% (2011 Census); estimated 82%+ now.
- **Urbanisation:** ~35% (Census 2011); slow by global standards (World 55%, China 65%).

Chapter 12 — Agriculture

12.1 3 cropping seasons

- **Kharif** (Jun–Oct, monsoon) — **rice, maize, bajra, jowar, ragi, cotton, sugarcane, tur, groundnut, soyabean, jute.**
- **Rabi** (Oct–Mar, winter) — **wheat (dominant), barley, gram, mustard, peas, lentil.**
- **Zaid** (Mar–Jun, summer) — watermelon, cucumber, muskmelon, fodder.

12.2 Crop → Largest producer state

Crop	Largest producer
Rice	West Bengal
Wheat	Uttar Pradesh
Pulses	Madhya Pradesh
Sugarcane	Uttar Pradesh ("sugar bowl")
Cotton	Gujarat
Tea	Assam
Coffee	Karnataka
Jute	West Bengal
Rubber	Kerala
Oilseeds	Madhya Pradesh / Rajasthan
Soyabean	Madhya Pradesh

Crop	Largest producer
Groundnut	Gujarat
Silk	Karnataka (mulberry)
Spices	Kerala / Gujarat
Coconut	Kerala / TN

12.3 Revolutions

- **Green Revolution** (1960s–70s, Norman Borlaug + M.S. Swaminathan) — wheat + rice in Punjab + Haryana + W. UP.
- **White Revolution** / Operation Flood (1970s–90s, Verghese Kurien, Amul) — milk; India now world's #1 milk producer.
- **Yellow Revolution** — oilseeds.
- **Blue Revolution** — fishery.
- **Pink Revolution** — meat / poultry.
- **Silver Revolution** — eggs.
- **Rainbow Revolution** — holistic agri growth.

12.4 Exam hooks

- India is world's **#1 in MILK, PULSES, JUTE, COTTON, SPICES, BANANA, MANGO**.
- India is **#2 in rice, wheat, sugarcane, tea, fish, onion, potato, groundnut**.
- **NORMAN BORLAUG** — Father of Green Revolution globally; **M.S. SWAMINATHAN** — Father of Indian Green Rev.
- **White Revolution** = Amul / Operation Flood — Verghese Kurien.

Chapter 13 — Minerals + Energy

13.1 Minerals — state-wise leaders

Mineral	Top state
Iron ore	Odisha (~55%)
Coal	Jharkhand (followed by Odisha, CG)
Bauxite (Al)	Odisha
Manganese	Madhya Pradesh
Copper	MP / Rajasthan
Mica	Andhra Pradesh / Jharkhand
Gold	Karnataka (Kolar, now closed; Hutti active)

Mineral	Top state
Uranium	Jharkhand (Jaduguda — only operating)
Diamond	Madhya Pradesh (Panna — only)
Gypsum	Rajasthan
Salt	Gujarat

13.2 Energy — India's mix (2024)

- **Total installed capacity ~452 GW.**
- **Non-fossil ~46%** (nearing 50% target).
- **Coal-based thermal: ~53%.**
- **Solar: ~93 GW.**
- **Wind: ~47 GW.**
- **Large hydro: ~47 GW.**
- **Nuclear: ~7.5 GW.**
- Target — **500 GW non-fossil by 2030; Net Zero by 2070.**

Nuclear plants: - **Tarapur** (Maharashtra) — India's oldest (1969). - **Rawatbhata** (Rajasthan), **Kakrapar** (Gujarat), **Kaiga** (Karnataka), **Kudankulam** (TN — largest, Russian VVER), **Kalpakkam** (TN — PFBR commissioned 2024), **Narora** (UP).

Chapter 14 — Transport + Communication

14.1 Road + rail

- **Road:** India has ~63 lakh km road network — **2nd largest globally.**
- **National Highways:** ~1.5 lakh km (post-Bharatmala additions).
- **Golden Quadrilateral** (NHAI, begun 1999–2000, Vajpayee) — 5,846 km; connects Delhi-Mumbai-Chennai-Kolkata.
- **North-South + East-West Corridors** — 7,300 km.
- **Railways:** ~68,000 route-km — Asia's largest rail network; **4th largest globally.**
- **First railway** — 16 April 1853, **Bombay to Thane** (34 km), Lord Dalhousie.
- **Zones** — currently **18 zones.**
- **Vande Bharat Express** — indigenous semi-high-speed train (since 2019).

14.2 Air + water

- **Major airports** — ~140 operational; 35+ international.
- **Busiest** — Delhi IGI, Mumbai, Bengaluru, Chennai, Hyderabad, Kolkata.

- **Ports** — 13 major + 200+ minor. **JNPT Mumbai** (container) + **Mundra** (private, Adani) + **Paradip** (Odisha) + **Kandla** (Gujarat).

14.3 Exam hooks

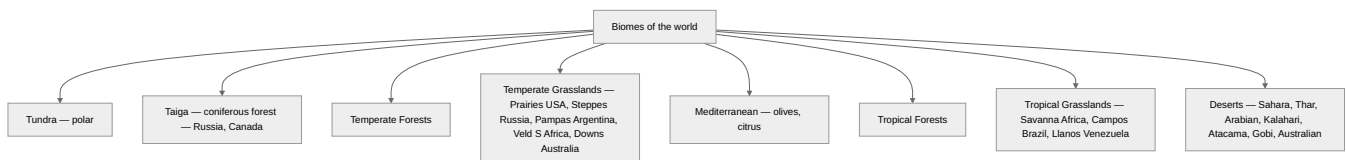
- First railway = **1853, Bombay-Thane**.
 - **Indian Railways** = world's 4th largest, Asia's largest network.
 - **Golden Quadrilateral** = Delhi-Mumbai-Chennai-Kolkata.
 - **Busiest airport** = Delhi IGI.
 - India's biggest port by cargo = **Mundra** (private).
-

Part F — World

Chapter 15 — Continents + Countries + Oceans (key facts)

- **7 continents** — Asia, Africa, N America, S America, Antarctica, Europe, Australia (by area).
- **Largest country** by area — Russia (~17.1 mn km²).
- **Largest by population** — **India** (overtook China, Apr 2023).
- **Longest river** — **Nile** (6,650 km) OR **Amazon** (debated) — standard answer is Nile; by water volume Amazon #1.
- **Largest lake** — Caspian Sea (by area); Baikal (by volume + depth).
- **Largest desert** — Antarctica (polar); Sahara is largest hot desert (9.2 mn km², across 11 countries).
- **Highest peak** — Mt Everest (8,848 m).
- **Longest mountain range (land)** — Andes (7,000 km); longest overall = Mid-Atlantic Ridge (underwater).
- **Largest ocean** — Pacific.
- **Deepest point** — Challenger Deep, Mariana Trench, 10,984 m.

Chapter 16 — Biomes (Natural Vegetation)



16.1 Exam hooks

- **Prairies** = N America grasslands; **Pampas** = Argentina; **Steppes** = Russia; **Veld** = S Africa; **Downs** = Australia; **Llanos** = Venezuela
- Colombia; **Campos** = Brazil; **Savanna** = Africa.

Appendices

Appendix A — India's 28 states + 8 UTs (capital shortcut)

State	Capital	State	Capital
AP	Amaravati (de facto)	Maharashtra	Mumbai
Arunachal Pradesh	Itanagar	Manipur	Imphal
Assam	Dispur (Guwahati)	Meghalaya	Shillong
Bihar	Patna	Mizoram	Aizawl
Chhattisgarh	Raipur	Nagaland	Kohima
Goa	Panaji	Odisha	Bhubaneswar
Gujarat	Gandhinagar	Punjab	Chandigarh
Haryana	Chandigarh	Rajasthan	Jaipur
Himachal	Shimla	Sikkim	Gangtok
Jharkhand	Ranchi	Tamil Nadu	Chennai
Karnataka	Bengaluru	Telangana	Hyderabad
Kerala	Thiruvananthapuram	Tripura	Agartala
MP	Bhopal	UP	Lucknow
Maharashtra	Mumbai	Uttarakhand	Dehradun / Gairsain (summer)
WB	Kolkata		

UTs (8): Delhi, Puducherry, J&K (3 with legislature); Chandigarh, A&N Islands, Lakshadweep, Dadra-Nagar-Haveli+Daman-Diu, Ladakh (5 without).

Appendix B — Mnemonic compendium

- **MVEMJSUN** → planets.
- **FED-RIVE** → IVC decline (shared with History).
- **Ring of Fire** → 81% of earthquakes.
- **Arabian Sea + Bay of Bengal** → two branches of SW monsoon.
- **Mawsynram > Cherrapunji** → wettest places.
- **Khadar vs Bhangar** → new vs old alluvium.
- **GR MaCH JH BeTrMi** → 8 Tropic-of-Cancer states.
- **"Western disturbances"** → N India winter rain.

- **"Nawab-Wilma"** cyclone-type names rotate in Indian Ocean basin.

Appendix C — Cheatsheet (1 page)

- **IST** — 82°30'E Mirzapur; UTC+5:30.
- **Total area** — 3.287 Mkm² (7th largest); coastline 7,517 km; land border 15,107 km.
- **Longest land border** — Bangladesh (4,096 km).
- **Shortest** — Afghanistan (106 km, via PoK).
- **Himalayas**: Shiwalik → Lesser → Greater → Trans; Kanchenjunga highest in India.
- **Rivers**: Ganga-Yamuna largest N; Godavari longest entirely in India; Narmada + Tapi flow WEST.
- **Monsoon onset** — Kerala 1 June; covers whole country by 15 July.
- **Wettest place** — Mawsynram (Meghalaya).
- **Tropic of Cancer** crosses **8 states**.
- **Alluvial soil** most widespread; black soil best for cotton.
- **Largest iron-ore producer state** — Odisha.
- **Largest wheat** — UP; largest rice — WB.
- Nuclear plants — Tarapur (oldest), Kudankulam (largest), Kalpakkam (fast breeder 2024).
- India — most populous since **Apr 2023** (~144 cr).
- Non-fossil target — **500 GW by 2030; Net Zero by 2070**.
- Golden Quadrilateral — Delhi-Mumbai-Chennai-Kolkata.

Parts C (oceans in detail), world countries' capitals, world cities + resources will be added in subsequent revisions of this file.

PART X — STATES OF INDIA — COMPLETE TABLE

State	Capital	Formed	Lok Sabha seats	Rajya Sabha	Notable
Andhra Pradesh	Amaravati (de facto Vizag/Kurnool tri-capital)	1 Nov 1956 (re-org 2 Jun 2014)	25	11	Telugu speaking
Arunachal Pradesh	Itanagar	20 Feb 1987 (UT 1972)	2	1	"Land of the rising sun"
Assam	Dispur	26 Jan 1950	14	7	Brahmaputra valley
Bihar	Patna	26 Jan 1950	40	16	Magadha heritage
Chhattisgarh	Raipur	1 Nov 2000	11	5	Carved out of MP
Goa	Panaji	30 May 1987	2	1	Smallest state by area
Gujarat	Gandhinagar	1 May 1960	26	11	Coastline 1,600 km (longest of any state)
Haryana	Chandigarh (shared with Punjab)	1 Nov 1966	10	5	Carved out of Punjab
Himachal Pradesh	Shimla (winter Dharamshala)	25 Jan 1971	4	3	Hill state
Jharkhand	Ranchi	15 Nov 2000	14	6	Mineral rich
Karnataka	Bengaluru	1 Nov 1956	28	12	IT capital
Kerala	Thiruvananthapuram	1 Nov 1956	20	9	Highest literacy
Madhya Pradesh	Bhopal	1 Nov 1956	29	11	Largest state (after split)
Maharashtra	Mumbai (winter Nagpur)	1 May 1960	48	19	Largest LS share
Manipur	Imphal	21 Jan 1972	2	1	Loktak Lake

State	Capital	Formed	Lok Sabha seats	Rajya Sabha	Notable
Meghalaya	Shillong	21 Jan 1972	2	1	"Abode of clouds"; Mawsynram wettest
Mizoram	Aizawl	20 Feb 1987	1	1	Highest literacy in NE
Nagaland	Kohima	1 Dec 1963	1	1	16+ tribes
Odisha	Bhubaneswar	1 Apr 1936	21	10	Cyclone-prone
Punjab	Chandigarh (shared)	1 Nov 1966	13	7	5 - river land
Rajasthan	Jaipur	30 Mar 1949	25	10	Largest state by area
Sikkim	Gangtok	16 May 1975 (22nd state)	1	1	Khangchendzonga
Tamil Nadu	Chennai	26 Jan 1950	39	18	Coromandel coast
Telangana	Hyderabad	2 Jun 2014 (29th state)	17	7	Carved from AP
Tripura	Agartala	21 Jan 1972	2	1	3rd smallest
Uttar Pradesh	Lucknow	26 Jan 1950	80	31	Most populous; max LS + RS seats
Uttarakhand	Dehradun (winter); Gairsain (summer 2020)	9 Nov 2000	5	3	Devbhoomi
West Bengal	Kolkata	26 Jan 1950	42	16	Sundarbans

8 Union Territories

UT	Capital	Special
Delhi (NCT)	New Delhi	LS 7, RS 3; Art 239AA; LG + CM
J&K	Srinagar (summer) / Jammu (winter)	LS 5, RS 4; UT since 31 Oct 2019; LG + CM
Ladakh	Leh + Kargil	LS 1; UT since 31 Oct 2019; LG only

UT	Capital	Special
Puducherry	Puducherry	LS 1, RS 1; LG + CM
Chandigarh	Chandigarh	shared capital Punjab+Haryana
Andaman & Nicobar	Port Blair (renamed Sri Vijaya Puram, 2024)	LS 1; LG only
Dadra & Nagar Haveli + Daman & Diu	Daman	merged 26 Jan 2020
Lakshadweep	Kavaratti	smallest UT; LS 1

Memory peg. UP = max LS (80) + RS (31). Sikkim, Mizoram, Nagaland = 1 LS each. Goa & Sikkim = smallest by area (Goa) / 2nd-smallest (Sikkim).



PART Y — MOUNTAINS, RIVERS, LAKES, SOILS

Major Indian mountains + peaks

Range	Highest peak	State	Notes
Greater Himalayas (Himadri)	Kanchenjunga 8,586 m	Sikkim	Highest in India; 3rd in world
Trans-Himalaya (Karakoram)	K2 (Godwin Austen) 8,611 m	PoK / Gilgit	World's 2nd; not in present India
Greater Himalayas	Nanda Devi 7,816 m	Uttarakhand	2nd in India proper
Lesser Himalayas (Himachal)	Nag Tibba	UP/HP	Pir Panjal, Dhauladhar, Mahabharat
Outer Himalayas (Shivaliks)	—	foothills	Bhabar, Terai zone
Eastern Himalayas	Namcha Barwa 7,782 m	Arunachal	
Aravalli	Guru Shikhar 1,722 m	Mt. Abu, Rajasthan	Oldest fold mountains in world
Vindhyas	Kalumar 752 m	MP	
Satpura	Dhupgarh 1,350 m	Pachmarhi, MP	
Western Ghats (Sahyadri)	Anamudi 2,695 m	Kerala	UNESCO; biodiversity hotspot
Eastern Ghats	Arma Konda 1,680 m	AP	discontinuous
Nilgiris	Doddabetta 2,637 m	TN	meeting of Western & Eastern Ghats
Annamalai Hills	Anamudi	Kerala/TN	
Cardamom Hills	Agastya Mala	Kerala	
Garo-Khasi-Jaintia	Shillong Peak 1,961 m	Meghalaya	Mawsynram + Cherrapunji
Patkai-Bum	—	Nagaland-Manipur-Mizoram-Arunachal	NE border

Major Indian rivers

River	Length (km)	Origin	Mouth	Tributaries (key)
Indus (Sindhu)	3,180 (320 in India)	Mansarovar (Tibet)	Arabian Sea (Pakistan)	Jhelum, Chenab, Ravi, Beas, Sutlej
Ganga	2,525	Gangotri (Bhagirathi + Alaknanda meet at Devprayag)	Bay of Bengal (Sundarbans)	Yamuna, Son, Ramganga, Gomti, Ghaghra, Gandak, Kosi, Damodar (RHS = right-hand-side: Yamuna, Son; LHS: rest)
Yamuna	1,376	Yamunotri	Allahabad (joins Ganga)	Chambal, Sind, Betwa, Ken
Brahmaputra	2,900 (916 in India)	Mansarovar (Tibet — "Tsangpo")	Bay of Bengal (Bangladesh — joins Ganga as Padma)	Subansiri, Manas, Lohit, Dhansiri, Teesta
Godavari	1,465	Triambak (MH)	Bay of Bengal (AP)	Penganga, Wardha, Indravati, Sabari — largest peninsular river
Krishna	1,400	Mahabaleshwar (MH)	Bay of Bengal (AP)	Bhima, Tungabhadra, Koyna, Musi, Malaprabha, Ghatprabha
Cauvery	800	Talakaveri (KA, Brahmagiri)	Bay of Bengal (Poompuhar, TN)	Hemavati, Kabini, Bhavani, Amaravati
Mahanadi	858	Sihawa (Chhattisgarh)	Bay of Bengal (Odisha)	Hirakud Dam (longest dam)
Narmada	1,312	Amarkantak (MP)	Arabian Sea (Bharuch, GJ)	flows through rift valley
Tapi (Tapti)	724	Multai (MP)	Arabian Sea (Surat, GJ)	flows in rift valley
Sabarmati	371	Aravalli (Rajasthan)	Gulf of Khambat	Ahmedabad
Mahi	583	Vindhya (MP)	Gulf of Khambat	crosses Tropic of Cancer twice
Subarnarekha	395	Chota Nagpur (Jharkhand)	Bay of Bengal	small but significant
Periyar	244	Sivagiri (Kerala)	Arabian Sea	longest river of Kerala

River	Length (km)	Origin	Mouth	Tributaries (key)
Luni	495	Aravalli	Rann of Kachchh	only river of Thar; ends in marsh
Damodar	592	Chota Nagpur	Hooghly (WB)	"Sorrow of Bengal" earlier
Teesta	414	Sikkim	joins Brahmaputra (Bangladesh)	

Peninsular rivers flowing west to Arabian Sea: Narmada, Tapi, Sabarmati, Mahi, Periyar, Luni — REST flow east.

Lakes (largest / famous)

Lake	State	Type / note
Wular	J&K	Largest freshwater lake in India
Dal	J&K	Famous tourist lake
Chilika	Odisha	Largest brackish/saltwater lake in India ; Ramsar
Pulicat	AP/TN	2nd largest brackish
Vembanad	Kerala	Longest lake in India (96 km); Ramsar
Sambhar	Rajasthan	Largest inland salt lake
Lonar	MH	Crater lake (meteoric)
Loktak	Manipur	Floating phumdis; Keibul Lamjao only floating NP
Hussain Sagar	Telangana	Buddha statue, Hyderabad
Pangong Tso	Ladakh	High-altitude saline lake; partially in China
Tso Moriri	Ladakh	High altitude (4,522 m)
Kolleru	AP	Largest freshwater lake in AP
Bhopal Upper Lake (Bada Talab)	MP	Built by Raja Bhoj (11c)
Nakki	Mt. Abu, Rajasthan	Hill lake
Naini	Uttarakhand	Crescent-shaped
Roopkund	Uttarakhand	"Skeleton lake"
Khecheopalri	Sikkim	Sacred to Buddhists + Hindus

Indian soils (8 types)

Soil	Region	Crops	Note
Alluvial	Indo - Gangetic plain	rice, wheat, sugarcane	Most widespread (~40% land); fertile
Black (Regur)	Deccan trap (MH, MP, GJ, KA)	cotton , sugarcane, jowar	Self-ploughing, retains moisture; basaltic origin
Red & Yellow	Eastern + S. Deccan	groundnut, millets	Iron oxide
Laterite	Western Ghats, NE, Odisha	tea, coffee, cashew	Heavy rainfall + alternate wet-dry
Arid (Desert)	Rajasthan	barley, millets, gram	Saline; needs irrigation
Saline & Alkaline	Punjab, Haryana, Rann of Kachchh	tolerant crops	Excess salts
Peaty & Marshy	Kerala backwaters, Sundarbans	rice	High organic matter
Forest & Mountain	Himalayas, Western Ghats	apples, fruits	Variable depth



PART Z — NATIONAL PARKS, TIGER + BIOSPHERE RESERVES

Tiger reserves (major — 50+ in India)

Reserve	State	Notes
Jim Corbett	Uttarakhand	India's first NP (1936); first Tiger Project (1973)
Bandhavgarh	MP	Highest tiger density
Kanha	MP	Inspired Kipling's Jungle Book; barasingha
Pench	MP/Maharashtra	Mowgli land
Satpura	MP	
Panna	MP	River Ken
Tadoba - Andhari	Maharashtra	Largest in MH
Melghat	Maharashtra	First tiger reserve in MH
Sariska	Rajasthan	Aravallis
Ranthambore	Rajasthan	Famous Machhli tigress
Mukundra Hills	Rajasthan	
Ramgarh Vishdhari	Rajasthan	52nd, 2022
Sundarbans	West Bengal	UNESCO; mangrove tigers
Buxa	West Bengal	
Manas	Assam	UNESCO; Indo - Bhutan
Nameri	Assam	
Kaziranga	Assam	UNESCO; one - horned rhino
Pakke	Arunachal	Hornbill
Namdapha	Arunachal	
Dampa	Mizoram	
Bhadra	Karnataka	
Bandipur	Karnataka	First "Project Tiger"
Nagarhole (Rajiv Gandhi)	Karnataka	Kabini

Reserve	State	Notes
Anshi	Karnataka	
BRT (Biligiri)	Karnataka	Recent
Periyar	Kerala	Boating
Parambikulam	Kerala	
Mudumalai	TN	Nilgiris
Kalakkad - Mundanthurai	TN	Southernmost
Sathyamangalam	TN	Veerappan land
Anamalai (Indira Gandhi)	TN	
Mukurthi	TN	Nilgiri tahr
Nagarjunasagar - Srisailem	AP-TG	Largest tiger reserve in India by area
Amrabad	Telangana	
Achanakmar	Chhattisgarh	
Indravati	Chhattisgarh	
Udanti - Sitanadi	Chhattisgarh	
Palamau	Jharkhand	One of the first 9 (1973)
Similipal	Odisha	
Satkosia	Odisha	
Dudhwa	UP	Bardia border
Pilibhit	UP	UNESCO
Amangarh	UP	
Valmiki	Bihar	
Kamlang	Arunachal	
Orang	Assam	Smallest
Veerangana Durgavati	MP	54th (2023)
Dholpur - Karauli	Rajasthan	55th (2024)
Rani Durgavati / Madhav	MP	proposed 2025

All Biosphere reserves (18, with UNESCO MAB status flagged)

#	Reserve	State	Year	UNESCO MAB
1	Nilgiri	TN/KE/KA	1986	✓ (2000)
2	Nanda Devi	UK	1988	✓ (2004)
3	Nokrek	Meghalaya	1988	✓ (2009)
4	Manas	Assam	1989	—
5	Sundarbans	WB	1989	✓ (2001)
6	Gulf of Mannar	TN	1989	✓ (2001)
7	Great Nicobar	A&N	1989	✓ (2013)
8	Similipal	Odisha	1994	✓ (2009)
9	Dibru - Saikhowa	Assam	1997	—
10	Dehang - Debang	Arunachal	1998	—
11	Pachmarhi	MP	1999	✓ (2009)
12	Khangchendzonga	Sikkim	2000	✓ (2018)
13	Agasthyamalai	KE/TN	2001	✓ (2016)
14	Achanakmar - Amarkantak	CG/MP	2005	✓ (2012)
15	Kachchh	Gujarat	2008	—
16	Cold Desert	HP	2009	—
17	Seshachalam	AP	2010	—
18	Panna	MP	2011	✓ (2020)

Ramsar sites — selected key (India had 80+ as of 2024)

- **Chilika Lake** (Odisha) — first Ramsar site of India (1981); largest brackish water lake.
- **Keoladeo NP / Bharatpur** (Rajasthan) — UNESCO + Ramsar; Siberian crane.
- **Wular Lake** (J&K) — largest freshwater.
- **Loktak Lake** (Manipur) — Keibul Lamjao floating NP.
- **Sundarban Wetland** (WB) — largest in India.
- **Vembanad-Kol Wetland** (Kerala) — longest lake of India.
- **Sambhar Lake** (Rajasthan) — largest salt lake.
- **Bhitarkanika Mangroves** (Odisha) — saltwater crocodile.
- **Hokera Wetland** (J&K).
- **Renuka Wetland** (HP).

- **Surinsar-Mansar** (J&K).
 - **Asan Conservation Reserve** (UK).
 - **Sandi Bird Sanctuary, Saman, Samaspur** (UP).
 - **Bhoj Wetland** (MP) — Bhopal lakes.
 - **Deepor Beel** (Assam).
 - **Point Calimere** (TN).
-

PART AA — WORLD GEOGRAPHY (high-yield)

Continents — quick facts

Continent	Area rank	Population rank	Largest country	Smallest
Asia	1	1	Russia (Asia + Europe) / China	Maldives
Africa	2	2	Algeria	Seychelles
North America	3	4	Canada	St. Kitts & Nevis
South America	4	5	Brazil	Suriname
Antarctica	5	—	(no permanent pop)	—
Europe	6	3	Russia (Europe)	Vatican
Australia / Oceania	7	6	Australia	Nauru

World — longest / highest / largest (memorise)

Record	Holder
Highest mountain	Mt. Everest (Sagarmatha / Chomolungma) — 8,848.86 m, Nepal-China
2nd highest	K2 (Godwin Austen) — 8,611 m
Longest river	Nile (Africa) — 6,650 km
2nd longest	Amazon (S. America) — 6,400 km
Longest in Asia	Yangtze (China) — 6,300 km
Largest desert (overall)	Antarctica (cold desert)
Largest hot desert	Sahara (Africa) — 9.2 M km ²
Largest ocean	Pacific
Smallest ocean	Arctic
Deepest point	Mariana Trench (Challenger Deep) — 11,034 m
Largest country (area)	Russia — 17.1 M km ²
2nd	Canada
3rd	USA
4th	China

Record	Holder
7th	India — 3.28 M km ²
Most populous country	India (~144 cr, since Apr 2023); China 2nd
Smallest country (area + pop)	Vatican City
Largest lake (area)	Caspian Sea (saline)
Largest freshwater lake (area)	Lake Superior (USA - Canada)
Deepest lake	Lake Baikal (Russia) — 1,642 m
Largest island	Greenland
Largest archipelago	Indonesia (17,500+ islands)
Largest peninsula	Arabian Peninsula
Largest bay	Bay of Bengal
Largest river basin	Amazon
Largest delta	Sundarbans (Ganga - Brahmaputra)
Largest waterfall (height)	Angel Falls (Venezuela) — 979 m
Largest by volume	Inga Falls (DRC)
Niagara	USA - Canada border
Highest plateau	Tibetan Plateau ("roof of the world")
Largest plateau	Tibetan Plateau
Highest active volcano	Ojos del Salado (Chile)
Largest hot desert (single)	Sahara
Largest cold desert	Antarctica; (in Asia: Gobi)

Important world deserts

Desert	Location
Sahara	N. Africa
Arabian	Middle East
Gobi	Mongolia + China
Kalahari	Southern Africa
Patagonian	Argentina

Desert	Location
Great Victoria	Australia
Atacama	Chile (driest non-polar)
Thar	India + Pakistan
Karakum	Turkmenistan
Mojave	USA

Important rivers (world)

River	Continent	Empties into
Nile	Africa	Mediterranean
Amazon	S. America	Atlantic
Yangtze	Asia	E. China Sea
Mississippi-Missouri	N. America	Gulf of Mexico
Yenisei-Angara	Asia	Arctic Ocean
Yellow (Huang He)	China	Bohai Sea
Ob	Russia	Arctic
Parana	S. America	Atlantic
Congo	Africa	Atlantic
Volga	Russia	Caspian (longest in Europe)
Danube	Europe	Black Sea (passes most countries — 10)
Rhine	Europe	North Sea
Mekong	SE Asia	South China Sea
Murray-Darling	Australia	Indian Ocean

Mountain ranges (world)

Range	Continent	Highest peak
Andes	S. America	Aconcagua (Argentina) — 6,961 m; longest range
Rockies	N. America	Mt. Elbert (USA) — 4,401 m
Himalayas	Asia	Everest 8,848

Range	Continent	Highest peak
Alps	Europe	Mont Blanc 4,810
Atlas	Africa	Toubkal (Morocco)
Ural	Russia	divides Europe + Asia
Caucasus	Europe-Asia	Mt. Elbrus (highest in Europe)
Great Dividing Range	Australia	Kosciuszko

Straits (key)

- Bering Strait — Russia + Alaska (Pacific-Arctic)
- Strait of Gibraltar — Spain + Morocco (Atlantic-Mediterranean)
- Strait of Hormuz — Iran + Oman (Persian Gulf-Arabian Sea)
- Bab-el-Mandeb — Yemen + Djibouti (Red Sea-Gulf of Aden)
- Strait of Malacca — Indonesia + Malaysia (busiest)
- Palk Strait — India + Sri Lanka
- Adam's Bridge — chain of shoals India-SL
- 9° + 10° + 8° channels — Lakshadweep islands; 8° = between Minicoy + Maldives
- Cook Strait — North + South islands of NZ
- Bering Strait — international date line near here

Continents to know longitude/latitude:

- **Tropic of Cancer** (23.5° N) passes through India: Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, WB, Tripura, Mizoram (8 states).
- **Equator** (0°) passes through 13 countries (Ecuador, Colombia, Brazil, Gabon, Congo, DRC, Uganda, Kenya, Somalia, Indonesia, etc.).
- **IST** = +5:30 GMT, based on **82.5° E** (passes through Mirzapur, UP).
- **Indian Standard Meridian** = 82.5° E (Allahabad meridian).
- India lies between 8°4' N to 37°6' N latitude, 68°7' E to 97°25' E longitude.



PART AB — INDIAN CLIMATE + AGRICULTURE

Climate types (Köppen for India)

- **Aw** (Tropical Savanna) — most of peninsular India
- **As** (Tropical) — TN coast (winter rain, retreating monsoon)
- **Amw** (Monsoon — short dry season) — SW coast
- **Bshw** (Semi-arid steppe) — Rajasthan, MP, AP plateau
- **Bwhw** (Arid hot desert) — Thar (Rajasthan, Kachchh)
- **Cwg** (Humid subtropical, dry winter) — North India plain
- **Dfc** (Subarctic) — Himalayan high altitudes
- **E** (Polar/Tundra) — high Himalayas

Monsoon

- **SW Monsoon** (Jun-Sep) — 75% of annual rainfall; Arabian Sea + Bay of Bengal branches.
- **NE / Retreating Monsoon** (Oct-Dec) — TN, parts of AP coast.
- **Mawsynram** (Meghalaya) — wettest place (~11,872 mm/yr); Cherrapunji 2nd.
- **Rain-shadow regions** — leeward Western Ghats (eastern slope dry).

Agricultural revolutions

Revolution	About	Key person
Green	Wheat + rice (HYV)	M.S. Swaminathan ("father")
White	Milk (Operation Flood, AMUL)	Verghese Kurien
Blue	Fish + aquaculture	Hiralal Chaudhuri / Arun Krishnan
Yellow	Oilseeds	Sam Pitroda (Yellow rev tech)
Pink	Onions + prawns	Durgesh Patel
Brown	Cocoa, coffee, leather	—
Golden	Fruits + horticulture	Nirpakh Tutaj
Silver	Eggs + poultry	Indira Gandhi era
Black	Petroleum	—
Grey	Wool / fertiliser	—
Red	Tomato + meat	—
Round	Potato	—

Revolution	About	Key person
Evergreen	Sustainable agriculture	M.S. Swaminathan

Cropping seasons

- **Kharif** — Jun-Oct; rice, maize, jowar, bajra, cotton, soybean, groundnut.
- **Rabi** — Oct-Mar; wheat, barley, gram, mustard.
- **Zaid** — Mar-Jun; watermelon, cucumber, vegetables.

Largest producer (India — 2024)

Crop	State
Rice	West Bengal
Wheat	Uttar Pradesh
Sugarcane	UP
Cotton	Gujarat
Jute	West Bengal
Tea	Assam
Coffee	Karnataka
Coconut	Kerala
Spices	Kerala
Tobacco	AP
Pulses	MP
Oilseeds	Rajasthan
Maize	Karnataka
Bajra	Rajasthan
Jowar	Maharashtra
Banana	TN
Mango	UP
Apple	J&K → HP (post-2019)

PART AC — GEOGRAPHY TRAP-RECOGNITION CARDS

Trap	Where	How to spot
Largest state by area vs population	India	Area = Rajasthan; Population = UP
Coastline longest	India	Gujarat (1,600 km), then AP, TN
Tropic of Cancer states (8)	India	Gu - Ra - MP - CG - JH - WB - TR - MZ ("GRMP-CJ-WT-MZ")
Indus tributaries (5)	Punjab rivers	Jhelum, Chenab, Ravi, Beas, Sutlej
West-flowing peninsular rivers	Geography	Narmada, Tapi, Sabarmati, Mahi, Periyar, Luni — REST flow east
Largest brackish lake vs freshwater	Lakes	Brackish = Chilika; freshwater = Wular
Longest river of India	India	Ganga (2,525 km within India); Brahmaputra is longer overall but only 916 km in India
First NP of India	NP	Jim Corbett (1936)
First Tiger reserve	Project Tiger	Bandipur (1973), simultaneous Project Tiger
Highest peak in India proper	Peaks	Kanchenjunga 8,586 m (K2 in PoK)
Mawsynram vs Cherrapunji	Wettest	Mawsynram > Cherrapunji (both Meghalaya)
Smallest state	India	Goa (area); Sikkim (2nd); UTs not counted
Largest UT	UTs	Ladakh (since 2019)
Wettest desert?	World	Antarctica is largest desert (cold); Sahara largest hot

PART AD — GEOGRAPHY MINI-MOCK (25 Questions · 25 Minutes)

Set a timer. No looking back. Mark your answers and check the key at the end.

1. Which is the largest state of India by area?
(a) Madhya Pradesh (b) Maharashtra (c) Rajasthan (d) Uttar Pradesh
2. The wettest place in India is:
(a) Cherrapunji (b) Mawsynram (c) Agumbe (d) Munnar
3. Which river flows westward into the Arabian Sea?
(a) Godavari (b) Krishna (c) Narmada (d) Cauvery
4. The Standard Meridian of India (82.5°E) passes through:
(a) Allahabad (b) Mirzapur (c) Varanasi (d) Lucknow
5. India's first National Park was:
(a) Bandipur (b) Kanha (c) Jim Corbett (d) Gir
6. Which is the only floating National Park in India?
(a) Dibru-Saikhowa (b) Keibul Lamjao (c) Kaziranga (d) Manas
7. Black soil (Regur) is ideal for growing which crop?
(a) Wheat (b) Rice (c) Cotton (d) Jute
8. The Tropic of Cancer passes through how many Indian states?
(a) 6 (b) 7 (c) 8 (d) 9
9. Which state has the longest coastline in India?
(a) Andhra Pradesh (b) Maharashtra (c) Tamil Nadu (d) Gujarat
10. India shares its longest international border with:
(a) China (b) Pakistan (c) Bangladesh (d) Nepal
11. The highest peak in India is:

(a) K2 (b) Kanchenjunga (c) Nanda Devi (d) Mt. Everest

12. Laterite soil is characterised by:

(a) High fertility and calcium (b) Leaching of nutrients in heavy rain (c) High nitrogen content (d) Found mainly in Indo-Gangetic plains

13. The Zoji La pass connects:

(a) Ladakh to Tibet (b) Srinagar to Leh (c) Manali to Leh (d) Sikkim to Tibet

14. Which among the following is a brackish water lake?

(a) Wular (b) Dal (c) Chilika (d) Bhimtal

15. The Deccan Plateau is bounded on which side by the Eastern Ghats?

(a) West (b) North (c) East (d) South

16. The monsoon enters India first from which direction?

(a) Bay of Bengal, hitting Odisha coast first (b) Arabian Sea, hitting Kerala coast first (c) Bay of Bengal, hitting West Bengal first (d) Arabian Sea, hitting Gujarat first

17. The soil found in the Indo-Gangetic plains is:

(a) Black soil (b) Red and yellow soil (c) Laterite soil (d) Alluvial soil

18. India's largest freshwater lake is:

(a) Dal (b) Wular (c) Chilika (d) Loktak

19. Mica, the mineral used in electronics, is most abundantly found in:

(a) Rajasthan (b) Jharkhand (c) Chhattisgarh (d) Odisha

20. The Western Disturbances bring winter rain to:

(a) Kerala and Tamil Nadu (b) Punjab, Haryana, and Rajasthan (c) Odisha and Andhra Pradesh (d) Bihar and West Bengal

21. India's first Ramsar Wetland Site was:

(a) Keoladeo Ghana (b) Loktak (c) Wular (d) Chilika

22. The Sundarbans mangrove forest is shared between India and:

(a) Sri Lanka (b) Myanmar (c) Bangladesh (d) Bhutan

23. The Palk Strait separates India from:

(a) Sri Lanka (b) Maldives (c) Bangladesh (d) Indonesia

24. The Brahmaputra river is known as "Tsangpo" in:

(a) Bhutan (b) Bangladesh (c) Nepal (d) Tibet (China)

25. The Father of the Green Revolution in India is:

(a) Norman Borlaug (b) Verghese Kurien (c) M.S. Swaminathan (d) C. Subramaniam

Answer Key

1-c | 2-b | 3-c | 4-b | 5-c | 6-b | 7-c | 8-c | 9-d | 10-c | 11-b | 12-b | 13-b | 14-c | 15-c | 16-b | 17-d | 18-b | 19-a | 20-b | 21-d | 22-c | 23-a | 24-d | 25-c

Score: 23–25 = Excellent · 18–22 = Good · Below 18 = Re-read Part D (India: physical geography) and Part E (human/economic geography).

⚠ EXAM ALERT

****Error-log rule:**** For every question you got wrong, write one sentence explaining what you missed. Retest yourself on only those questions 3 days later — without looking at the solution. This single habit doubles what the mock teaches you.

Note for Q25: M.S. Swaminathan is the father of the Green Revolution in India; Norman Borlaug is the global father. Examiners ask both — do not confuse them.

Drill these tables. Geography past-paper patterns repeat year-over-year — these tables are 90%+ Geography mastery.

PART AE — INDIA "LONGEST / LARGEST / HIGHEST"

MASTER LIST

Examiners ask one of these every paper. Memorise this entire list cold.

Largest / longest / highest — INDIA

Geography

Record	Holder
Largest state (area)	Rajasthan (3.42 lakh km ²)
Smallest state (area)	Goa (3,702 km ²)
Largest UT (area)	Ladakh (since 2019)
Smallest UT (area)	Lakshadweep (32 km ²)
Most populous state	Uttar Pradesh (~24 cr)
Least populous state	Sikkim (~7 lakh)
Most populous UT	Delhi (~3 cr)
Highest population density	Bihar (1,106/km ²)
Lowest population density	Arunachal Pradesh (17/km ²)
Highest sex ratio (state)	Kerala (1,084 women / 1,000 men)
Lowest sex ratio (state)	Haryana (879)
Highest literacy state	Kerala (94%)
Lowest literacy state	Bihar (62%)
Longest coastline (state)	Gujarat (~1,600 km)
2nd longest coastline	Andhra Pradesh
Largest delta in India (also world)	Sundarbans (Ganga-Brahmaputra)
Highest peak in India	Kanchenjunga (8,586 m, Sikkim)
Highest peak in India proper (entirely Indian)	Nanda Devi (7,816 m, Uttarakhand)
Highest peak in S. India	Anamudi (2,695 m, Kerala)

Record	Holder
Highest mountain pass	Marsimik La (Ladakh, 5,582 m)
Highest battlefield in world	Siachen Glacier (~5,400 m)
Highest waterfall in India	Kunchikal Falls (Karnataka, 455 m)
2nd highest waterfall	Barehipani Falls (Odisha, 399 m)
Famous Jog Falls (Sharavathi)	Karnataka, 253 m
Highest motorable road	Umling La Pass (Ladakh, 19,300 ft)
Highest village	Komik (Spiti, HP)
Highest cricket ground	Chail (HP, 2,144 m)
Largest desert	Thar (Rajasthan)
Largest river island	Majuli (Assam, on Brahmaputra) — 2nd largest after Marajó (Brazil)
Largest fresh - water lake	Wular (J&K)
Largest brackish/salt water lake	Chilika (Odisha)
Largest inland salt lake	Sambhar (Rajasthan)
Longest lake	Vembanad (Kerala, 96 km)
Largest dam	Hirakud (Mahanadi, Odisha) — also longest earthen dam
Highest dam	Tehri Dam (Uttarakhand, 260.5 m)
Highest gravity dam	Bhakra Dam (Sutlej, HP-Punjab, 226 m)
Longest river within India	Ganga (~2,525 km in India)
Largest river by water volume	Brahmaputra
Longest tributary of Ganga	Yamuna
Largest river of South India	Godavari ("Dakshin Ganga", 1,465 km)
Largest river island (above ref)	Majuli
Highest plateau	Tibetan Plateau extension; Ladakh plateau within India
Largest plateau (India)	Deccan Plateau
Northernmost point	Indira Col (Siachen)
Southernmost point	Indira Point (Great Nicobar)
Easternmost point	Kibithu (Arunachal)

Record	Holder
Westernmost point	Guhar Moti (Gujarat)
Tropic of Cancer states (8)	Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, WB, Tripura, Mizoram

Wildlife / Environment

Record	Holder
First National Park	Jim Corbett (Uttarakhand, 1936)
First Tiger Reserve	Bandipur (Karnataka, 1973)
Largest National Park (area)	Hemis NP (Ladakh, 4,400 km ²)
Largest Tiger Reserve	Nagarjunasagar - Srisailem (AP+TG, 3,728 km ²)
Smallest National Park	South Button Island NP (Andamans, 0.03 km ²)
First Biosphere Reserve	Nilgiri (1986)
First Ramsar site	Chilika (Odisha, 1981)
Largest Biosphere Reserve	Gulf of Mannar
Largest mangrove forest	Sundarbans (West Bengal)
Wettest place	Mawsynram (Meghalaya, ~11,872 mm/yr)
2nd wettest	Cherrapunji (Sohra, Meghalaya)
Driest place	Jaisalmer (Rajasthan, ~150 mm/yr)
Hottest place recorded	Phalodi (Rajasthan, 51 °C, 2016)
Coldest inhabited place	Drass (Ladakh, -45 °C recorded)
Most polluted city (recurrent)	Delhi (NCR)

Infrastructure

Record	Holder
Largest port (cargo handled)	JNPT/Mumbai (Nhava Sheva)
Longest natural port	Vishakhapatnam
Busiest port	Mundra (private, Adani, GJ)
Longest river bridge	Bhupen Hazarika Setu (Lohit, Assam - Arunachal, 9.15 km)

Record	Holder
Longest sea bridge	Atal Setu (MTHL, Mumbai, 22 km, opened Jan 2024)
Longest road bridge over water	Atal Setu
Longest cable-stayed bridge	Sudarshan Setu / Gulf of Khambhat (in development)
Longest railway bridge	New Pamban Bridge (TN, inaugurated 6 Apr 2025 by PM Modi on Ram Navami; India's first vertical-lift railway sea bridge; 2.07 km, 99 spans, 72.5 m vertical-lift span; replaces Pamban Bridge 1914)
Highest railway bridge (world)	Chenab Rail Bridge (J&K, 359 m above river, Aug 2022 commissioned 2024)
Longest highway tunnel	Atal Tunnel (Rohtang, HP, 9.02 km, opened Oct 2020)
Longest railway platform	Hubballi (1,507 m, Karnataka — record since 2023)
Longest national highway	NH 44 (Srinagar to Kanyakumari, 4,112 km)
Densest road network	Kerala
Largest railway zone	Northern Railway
Smallest railway zone	North Eastern Railway
Highest railway station	Ghum (West Bengal, Darjeeling Himalayan Railway, 2,258 m)
Highest broad-gauge railway station	Chenab area, Reasi-Banihal section (J&K)
Largest stadium (capacity)	Narendra Modi Stadium (Motera, Ahmedabad, 1.32 lakh)
Largest cricket ground (older)	Eden Gardens (Kolkata)
Tallest statue	Statue of Unity (Kevadia, Gujarat, 182 m, Sardar Patel, 2018) — world's tallest
Largest religious structure (under construction)	Ram Mandir, Ayodhya (consecrated Jan 2024)
Largest temple (area)	Sri Ranganathaswamy Temple (Srirangam, TN, 156 acres)
Tallest temple gopuram	Murudeshwara (Karnataka)
Highest tower	Statue of Unity (182 m); next Kolkata Tower of Joy planned

Record	Holder
Tallest bridge tower (cable - stayed)	Signature Bridge (Delhi, 154 m, Yamuna)
Largest dome	Lotus Temple (Delhi, Bahai)
Largest church	Sé Cathedral (Goa)
Largest mosque	Jama Masjid (Delhi, Shah Jahan)
Oldest mountain range	Aravallis (Rajasthan)
Largest cave	Krem Liat Prah (Meghalaya, longest cave in India)

People / Government

Record	Holder
First President	Dr. Rajendra Prasad
First PM	Jawaharlal Nehru
First woman PM	Indira Gandhi
First woman President	Pratibha Patil (2007 - 12)
First tribal President	Droupadi Murmu (2022 -)
First Dalit President	K.R. Narayanan (1997 - 2002)
First woman Speaker	Meira Kumar (2009 - 14)
First Indian to space	Rakesh Sharma (1984, Soyuz T- 11)
First woman in space (Indian - origin)	Kalpana Chawla (1997)
First Indian to win Nobel	Rabindranath Tagore (1913, Literature)
First Indian woman Nobel	(Indian - origin) Mother Teresa (1979 Peace)
First Indian to summit Everest	Tenzing Norgay (with Hillary, 29 May 1953); first Indian citizen Avtar Singh Cheema (1965)
First Indian woman to summit Everest	Bachendri Pal (23 May 1984)
First Indian Olympic gold	Norman Pritchard (1900 Paris, 2 silver) — actually colonial; First independent Indian gold = Hockey 1948 (team)
First individual Olympic gold	Abhinav Bindra (2008 Beijing, shooting)

Record	Holder
First individual track Olympic gold	Neeraj Chopra (2020 Tokyo, javelin)
First chess World Champion (Indian)	Viswanathan Anand (2000-12)
Youngest chess World Champion	D Gukesh (Dec 2024, age 18)

Misc

Record	Holder
Largest postal network in world	India Post (1.55 lakh post offices)
Largest film industry	India (multiple — Bollywood + Tollywood + others)
Largest railway network in Asia (4th in world)	Indian Railways
Largest road network (3rd world)	India
Largest electricity-generating state	Maharashtra
Largest installed renewable capacity state	Rajasthan (solar + wind)
Largest milk producer state	Uttar Pradesh
Largest banana producer	Tamil Nadu
Largest mango producer	Uttar Pradesh
Largest tea producer	Assam
Largest coffee producer	Karnataka
Largest coconut producer	Kerala
Largest cotton producer	Gujarat
Largest sugarcane producer	Uttar Pradesh
Largest rice producer	West Bengal
Largest wheat producer	Uttar Pradesh
Largest jute producer	West Bengal
Largest oil refinery	Jamnagar (Gujarat — Reliance, world's largest single-location refinery)

Record	Holder
Largest port (container)	Mundra
Largest dam reservoir (capacity)	Indira Sagar (MP, Narmada)
Largest temple complex	Angkor Wat — Cambodia (NOT India); India's largest = Srirangam (TN)
Highest cricket strike rate ever	Sehwag / Rohit / Surya
First aircraft carrier	INS Vikramaditya (Russian - origin); first indigenous = INS Vikrant (2022)

PART AF — WORLD "LONGEST / LARGEST" QUICK-RECALL

Record	Holder
Tallest building	Burj Khalifa (Dubai, 828 m)
2nd tallest	Merdeka 118 (KL, 678 m)
Longest bridge	Danyang - Kunshan Grand Bridge (China, 164 km, rail)
Longest road bridge	Hong Kong–Zhuhai–Macau Bridge (55 km, sea-crossing)
Longest tunnel	Gotthard Base Tunnel (Switzerland, 57 km, rail)
Largest hotel	First World Hotel (Malaysia, 7,351 rooms)
Largest religious building	Angkor Wat (Cambodia, 162.6 ha)
Tallest statue	Statue of Unity (India, 182 m)
Tallest tower (free-standing)	Tokyo Skytree (634 m)
Largest stadium (capacity)	Narendra Modi Stadium (132,000); Rungrado May Day (Pyongyang, ~114,000)
Largest park	Northeast Greenland NP (972,000 km ²)
Largest reef system	Great Barrier Reef (Australia, 2,300 km)
Largest glacier (non-polar)	Siachen Glacier (India-Pak); Lambert Glacier (Antarctica) world's largest
Largest active volcano	Mauna Loa (Hawaii)
Largest crater	Vredefort (S. Africa, 300 km diameter); Chicxulub (Mexico, dinosaur extinction)
Largest delta	Sundarbans (India-Bangladesh)
Largest island	Greenland (2.16 M km ²)
Largest archipelago	Indonesia (17,500+ islands)
Largest peninsula	Arabian Peninsula
Highest mountain	Mt. Everest (8,848 m)
2nd	K2 (8,611 m)
Highest active volcano	Ojos del Salado (Chile)

Record	Holder
Highest waterfall (single drop)	Angel Falls (Venezuela, 979 m)
Largest waterfall (volume)	Inga Falls (DRC)
Largest river basin	Amazon
Longest river	Nile (Africa, 6,650 km)
2nd longest	Amazon (6,400 km)
Largest river by discharge	Amazon
Largest lake (area)	Caspian Sea (saline)
Largest freshwater lake	Lake Superior (USA-Canada)
Deepest lake	Baikal (Russia, 1,642 m)
Highest lake (navigable)	Titicaca (Bolivia-Peru, 3,812 m)
Largest desert	Antarctica (cold, 14 M km ²)
Largest hot desert	Sahara (9.2 M km ²)
Driest desert	Atacama (Chile)
Highest plateau	Tibetan Plateau ("Roof of the World")
Largest country (area)	Russia (17.1 M km ²)
2nd	Canada
3rd	USA
4th	China
7th	India
Smallest country	Vatican City (0.49 km ²)
Most populous country	India (~144 cr, since Apr 2023)
2nd	China
Most spoken language (native)	Mandarin Chinese
2nd	Spanish
3rd	English (most spoken total speakers)
Most spoken language in India	Hindi

Record	Holder
Largest tropical rainforest	Amazon (Brazil)
Largest coral reef	Great Barrier Reef
Largest hot desert	Sahara
Smallest ocean	Arctic
Largest ocean	Pacific
Deepest ocean point	Mariana Trench (Challenger Deep, 11,034 m)

Drill these "longest/largest/highest" lists till instant-recall. They alone account for 5-7 marks per paper.